Adam Czepielik

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Summary

Data Scientist with over 5 years of experience developing production-grade machine learning solutions across insurance and retail domains. Proven track record in delivering end-to-end data science projects from exploratory analysis through model deployment and monitoring.

Professional Experience

Data Scientist | Verisk Analytics | November 2020 – Present

International team, English working language, cross-continent collaborations

- **Insurance Model Fairness & Bias Mitigation (FairCheck):** Led implementation of first-to-market insurance pricing fairness review product; developed methodology and algorithmic solutions for bias detection and mitigation; collaborated with major US insurance company to ensure compliance with Colorado SB 21-169 | *DVC, polars, numpy, h2o, statsmodels, scipy*
- **GPU-Accelerated Domain Modeling Library:** Built domain-specific, GPU-accelerated modeling library; reduced model training time by 80% while lowering AWS infrastructure costs; implemented CI/CD pipelines with automated testing and deployment | *JAX, pyarrow, Equinox, GitLab CI/CD, poetry, mypy* | *Lean Six Sigma Green Belt Project (In progresss,expected completion in December 2025)*
- **Insurance Pricing Model Optimization (ISO Risk Analyzer):** Enhanced data preprocessing pipeline for core product with legacy codebase, achieving 30% speed improvement | *Luigi, DuckDB, h2o, PySpark*
- **Court Document Classification (Court Side):** Developed NLP classification system using BERT-family models in PyTorch to filter relevant court documents for expert review, improving balanced accuracy score over previous baseline model by 12%.
- **Data Integration & Migration:** Migrated pipeline from MS SQL Server to Spark on AWS EMR; integrated acquired company data using fuzzy matching techniques | *AWS Glue, Athena, pandas, geopandas*
- **Additional:** Maintainer of in-house Python tools package; developer of matplotlib branding theme; mentor in internal training program; POC development for local AI assistants using llama.cpp

Data Analyst | AVSystem | March 2019 – October 2020

Good Lood Team (ice cream retail network automation project) | Working closely with software developers, especially backend specialists | Agile/Scrum environment, Jira project management

- **Data Integration & Reporting:** Developed reporting tool integrating data from sales, warehouse, delivery, and loyalty program systems; built custom ETL package in R for data pipeline automation | *PostgreSQL, MongoDB, BigQuery, InfluxDB, SQLite, R, Python, AWS EC2*
- **Delivery Zone Optimization:** Provided analytical support for delivery zone delimitation project involving software simulations, real-world testing, and production deployment | *OSRM, Google OR-Tools, Docker, Grafana*
- **Demand Forecasting:** Implemented time series forecasting using Bayesian methods.

Technical Skills

- **Programming & Data Science:** Python (main), R, SQL
- **Notable Libraries:** jax, scikit-learn, PyTorch, h2o, scipy, statsmodels, PySpark, NumPy, pandas, geopandas, pyarrow, polars, luigi
- **Data Visualization:** Matplotlib, Seaborn, ggplot, Plotly, Grafana
- **MLOps & Deployment:** Docker, DVC, CI/CD (GitLab),
- **Databases:** PostgreSQL, MongoDB, BigQuery, InfluxDB, DuckDB, SQLite, MS SQL Server
- **Cloud:** AWS (EC2, S3, EMR, Glue, Athena), GCP (Compute Engine, App Engine, Artifact Registry)
- **Version Control & PM:** Git, GitLab, Jira
- **Methodologies:** Agile/Scrum, Lean Six Sigma (Green Belt candidate)

Education

MSc in Mathematics | Jagiellonian University, Kraków | 2014 – 2020
Erasmus Exchange: KU Leuven, Belgium (Fall 2018)

Selected Freelance and Personal Projects

Tram Delays Analysis (aczepielik.github.io/en/post/kraktram.en/)
End-to-end public transport analysis: API automation, web scraping, quantile regression modeling, interactive visualizations. Featured in local media and appreciated by transport specialists.

NGO Donor Analytics Dashboard
Hidden Markov Chain modeling for donor behavior patterns. Deployed using Docker, ShinyProxy, and Google Cloud.

Diophantus' Arithmetics Translation (PL, aczepielik.github.io/files/Arithmetica.pdf)
Translated fragments of *Arithmetics* by Diophantus of Alexandria from ancient Greek to Polish.